

Joins & Set Operations

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1. Creation of database

The screenshot displays the SQL Developer interface. On the left, the 'SCHEMAS' tree shows a list of databases: bankdb, bookflow_db, bookflow_week2_db, college_yt, practice, sql_clg, and sys. The main editor window shows two SQL statements: 'CREATE DATABASE IF NOT EXISTS joins;' and 'USE joins;'. The bottom output window shows the execution results of these statements.

#	Time	Action	Message
✓ 1	20:38:47	drop database bank_management	1 row(s) affected
✓ 2	20:44:48	CREATE DATABASE IF NOT EXISTS joins	1 row(s) affected
✓ 3	20:45:10	USE joins	0 row(s) affected

2. Create the tables

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- bankdb
- bookflow_db
- bookflow_week2_db
- college_yt
- practice
- sql_cdg
- sys

Administration Schemas

Information

No object selected

WEEK1* ddl WEEK2 SQL File 5* SQL File 6* SQL File 7*

1 CREATE DATABASE IF NOT EXISTS joins;

2 USE joins;

3

4 CREATE TABLE class (

5 id INT,

6 name VARCHAR(30)

7);

8 CREATE TABLE class_info (

9 id INT,

10 address VARCHAR(30)

11);

Output

Action Output

#	Time	Action	Message
1	20:38:47	drop database bank_management	1 row(s) affected
2	20:44:48	CREATE DATABASE IF NOT EXISTS joins	1 row(s) affected
3	20:45:10	USE joins	0 row(s) affected
4	20:46:10	CREATE TABLE class (id INT, name VA...	0 row(s) affected
5	20:46:18	CREATE TABLE class_info (id INT, addr...	0 row(s) affected

3. Insertion of data

File Edit View Query Database Server Tools Scripting Help

WEEK1* ddl WEEK2 SQL File 5* SQL File 6* SQL File 7

Limit to 1000 rows

SCHEMAS

Filter objects

- bankdb
- bookflow_db
- bookflow_week2_db
- college_yt
- practice
- sql_clg
- sys

Administration Schemas

Information

No object selected

Object Info Session

```

2 • USE joins;
3
4 • CREATE TABLE class (
5     id INT,
6     name VARCHAR(30)
7 );
8 • CREATE TABLE class_info (
9     id INT,
10    address VARCHAR(30)
11 );
12 • INSERT INTO class VALUES
13 (1, 'Sai Akhil'),
14 (2, 'Adhvith'),
15 (4, 'Kiran');
16
17 • INSERT INTO class_info VALUES
18 (1, 'HYDERABAD'),
19 (2, 'DELHI'),
20 (3, 'CHENNAI');

```

Output

Action Output

#	Time	Action	Message
✓ 2	20:44:48	CREATE DATABASE IF NOT EXISTS joins	1 row(s) affected
✓ 3	20:45:10	USE joins	0 row(s) affected
✓ 4	20:46:10	CREATE TABLE class (id INT, name V...	0 row(s) affected
✓ 5	20:46:18	CREATE TABLE class_info (id INT, ad...	0 row(s) affected
✓ 6	20:47:22	INSERT INTO class VALUES (1, 'Sai Akhil'),...	3 row(s) affected Record
✓ 7	20:47:34	INSERT INTO class_info VALUES (1, 'HYD...	3 row(s) affected Record

4. CROSS JOIN

CHEMAS

Filter objects

bankdb
bookflow_db
bookflow_week2_db
college_yt
practice
sql_clg
sys

Limit to 1000 rows

```

17 • INSERT INTO class_info VALUES
18   (1, 'HYDERABAD'),
19   (2, 'DELHI'),
20   (3, 'CHENNAI');
21 • SELECT *
22   FROM class
23   CROSS JOIN class_info;

```

Result Grid
Filter Rows:
Export:
Wrap Cell Content:

	id	name	id	address
▶	4	Kiran	1	HYDERABAD
	2	Adhvith	1	HYDERABAD
	1	Sai Akhil	1	HYDERABAD
	4	Kiran	2	DELHI
	2	Adhvith	2	DELHI
	1	Sai Akhil	2	DELHI
	4	Kiran	3	CHENNAI
	2	Adhvith	3	CHENNAI
	1	Sai Akhil	3	CHENNAI

Result 1 x

Output

Action Output

#	Time	Action	Message
✓ 3	20:45:10	USE joins	0 row(s) affected
✓ 4	20:46:10	CREATE TABLE class (id INT, name ...	0 row(s) affected
✓ 5	20:46:18	CREATE TABLE class_info (id INT, a...	0 row(s) affected
✓ 6	20:47:22	INSERT INTO class VALUES (1, 'Sai Akhil'...	3 row(s) affected Records: 3 Du
✓ 7	20:47:34	INSERT INTO class_info VALUES (1, 'HY...	3 row(s) affected Records: 3 Du
✓ 8	20:48:10	SELECT * FROM class CROSS JOIN class...	9 row(s) returned

Object Info
Session

Administration
Schemas

Information

No object selected

5. INNER JOIN

Navigator: WEEK1* ddl WEEK2 SQL File 5* SQL File 6* SQL

SCHEMAS

Filter objects

- bankdb
- bookflow_db
- bookflow_week2_db
- college_yt
- practice
- sql_clg
- sys

```

21 • SELECT *
22 FROM class
23 CROSS JOIN class_info;
24 • SELECT *
25 FROM class
26 INNER JOIN class_info
27 ON class.id = class_info.id;
--

```

Result Grid | Filter Rows: | Export: | Wrap Cell C

	id	name	id	address
▶	1	Sai Akhil	1	HYDERABAD
	2	Adhvith	2	DELHI

Administration Schemas

Information

No object selected

Result 2 x

Output

Action Output

#	Time	Action	Message
✓ 4	20:46:10	CREATE TABLE class (id INT, name ...	0 row(s) affected
✓ 5	20:46:18	CREATE TABLE class_info (id INT, a...	0 row(s) affected
✓ 6	20:47:22	INSERT INTO class VALUES (1, 'Sai Akhil'...	3 row(s) affected R
✓ 7	20:47:34	INSERT INTO class_info VALUES (1, 'HY...	3 row(s) affected R
✓ 8	20:48:10	SELECT * FROM class CROSS JOIN class...	9 row(s) returned
✓ 9	20:48:46	SELECT * FROM class INNER JOIN class_...	2 row(s) returned

Object Info Session

6. INNER JOIN — selected columns

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- bankdb
- bookflow_db
- bookflow_week2_db
- college_yt
- practice
- sql_clg
- sys

WEEK1* ddl WEEK2 SQL File 5* SQL File 6* SQL File 7* SQL File 8*

Limit to 1000 rows

```

25 FROM class
26 INNER JOIN class_info
27 ON class.id = class_info.id;
28 • SELECT class.name, class_info.address
29 FROM class
30 INNER JOIN class_info
31 ON class.id = class_info.id;

```

Result Grid

	name	address
▶	Sai Akhil	HYDERABAD
	Adhvith	DELHI

Export: Wrap Cell Content:

Administration Schemas

Information

No object selected

Object Info Session

Result 3 x

Output

Action Output

#	Time	Action	Message
✓ 5	20:46:18	CREATE TABLE class_info (id INT, a...	0 row(s) affected
✓ 6	20:47:22	INSERT INTO class VALUES (1, 'Sai Akhil'...	3 row(s) affected Records: 3 Duplicates: 0
✓ 7	20:47:34	INSERT INTO class_info VALUES (1, 'HY...	3 row(s) affected Records: 3 Duplicates: 0
✓ 8	20:48:10	SELECT * FROM class CROSS JOIN class...	9 row(s) returned
✓ 9	20:48:46	SELECT * FROM class INNER JOIN class_...	2 row(s) returned
✓ 10	20:49:17	SELECT class.name, class_info.address F...	2 row(s) returned

7. NATURAL JOIN

Navigator: WEEK1* ddl WEEK2 SQL File 5* SQL File 6* SQL File 7* SQL

SCHEMAS

Filter objects

- bankdb
- bookflow_db
- bookflow_week2_db
- college_yt
- practice
- sql_clg
- sys

Administration Schemas

Information

No object selected

Result Grid

id	name	address
1	Sai Akhil	HYDERABAD
2	Adhvith	DELHI

Result 4 ×

Output

Action Output

#	Time	Action	Message
6	20:47:22	INSERT INTO class VALUES (1, 'Sai Akhil'...	3 row(s) affected Records: 3 Duplicat
7	20:47:34	INSERT INTO class_info VALUES (1, 'HY...	3 row(s) affected Records: 3 Duplicat
8	20:48:10	SELECT * FROM class CROSS JOIN class...	9 row(s) returned
9	20:48:46	SELECT * FROM class INNER JOIN class...	2 row(s) returned
10	20:49:17	SELECT class.name, class_info.address F...	2 row(s) returned
11	20:49:50	SELECT * FROM class NATURAL JOIN cl...	2 row(s) returned

8. Add More Data for OUTER JOINS

WEEK1*

ddl

WEEK2

SQL File 5*

SQL File 6*

SQL

Limit to 1000 rows

23

CROSS JOIN class_info;

24

SELECT *

25

FROM class

26

INNER JOIN class_info

27

ON class.id = class_info.id;

28

SELECT class.name, class_info.address

29

FROM class

30

INNER JOIN class_info

31

ON class.id = class_info.id;

32

SELECT *

33

FROM class

34

NATURAL JOIN class_info;

35

INSERT INTO class VALUES

36

(5, 'Nikhil');

37

38

INSERT INTO class_info VALUES

39

(7, 'BENGALURU');

40

(8, 'PUNE');

41

administration

Schemas

information

No object selected

Object Info

Session

Output

Action Output

#	Time	Action	Message
8	20:48:10	SELECT * FROM class CROSS JOIN class...	9 row(s) returned
9	20:48:46	SELECT * FROM class INNER JOIN class_...	2 row(s) returned
10	20:49:17	SELECT class.name, class_info.address F...	2 row(s) returned
11	20:49:50	SELECT * FROM class NATURAL JOIN cl...	2 row(s) returned
12	20:50:21	INSERT INTO class_info VALUES (7, 'BEN...	2 row(s) affected R
13	20:50:25	INSERT INTO class VALUES (5, 'Nikhil')	1 row(s) affected

9. LEFT OUTER JOIN

CHEMAS

Filter objects

- bankdb
- bookflow_db
- bookflow_week2_db
- college_yt
- practice
- sql_clg
- sys

Limit to 1000 rows

38

•

INSERT INTO class_info VALUES

39

(7, 'BENGALURU'),

40

(8, 'PUNE');

41

•

SELECT *

42

FROM class

43

LEFT OUTER JOIN class_info

44

ON class.id = class_info.id;

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	id	name	id	address
▶	1	Sai Akhil	1	HYDERABAD
	2	Adhvith	2	DELHI
	4	Kiran	NULL	NULL
	5	Nikhil	NULL	NULL

administration

Schemas

information

No object selected

Result 5 ×

Output

Action Output

#	Time	Action	Message
✓ 9	20:48:46	SELECT * FROM class INNER JOIN class_...	2 row(s) returned
✓ 10	20:49:17	SELECT class.name, class_info.address F...	2 row(s) returned
✓ 11	20:49:50	SELECT * FROM class NATURAL JOIN cl...	2 row(s) returned
✓ 12	20:50:21	INSERT INTO class_info VALUES (7, 'BEN...	2 row(s) affected Records: 2 Duplicates: 0 ...
✓ 13	20:50:25	INSERT INTO class VALUES (5, 'Nikhil')	1 row(s) affected
✓ 14	20:52:04	SELECT * FROM class LEFT OUTER JOIN...	4 row(s) returned

10. LEFT JOIN

SCHEMAS

Filter objects

- bankdb
- bookflow_db
- bookflow_week2_db
- college_yt
- practice
- sql_cdg
- sys

Administration

Schemas

Information

No object selected

Limit to 1000 rows

43 LEFT OUTER JOIN class_info

44 ON class.id = class_info.id;

45 SELECT *

46 FROM class

47 LEFT JOIN class_info

48 ON class.id = class_info.id

49 WHERE class_info.id IS NULL;

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	id	name	id	address
▶	4	Kiran	NULL	NULL
	5	Nikhil	NULL	NULL

Result 6 ×

Output

Action Output

#	Time	Action	Message
✓ 10	20:49:17	SELECT class.name, class_info.address F...	2 row(s) returned
✓ 11	20:49:50	SELECT * FROM class NATURAL JOIN cl...	2 row(s) returned
✓ 12	20:50:21	INSERT INTO class_info VALUES (7, 'BEN...	2 row(s) affected Records: 2 Duplicates
✓ 13	20:50:25	INSERT INTO class VALUES (5, 'Nikhil')	1 row(s) affected
✓ 14	20:52:04	SELECT * FROM class LEFT OUTER JOIN...	4 row(s) returned
✓ 15	20:52:51	SELECT * FROM class LEFT JOIN class_in...	2 row(s) returned

11. RIGHT OUTER JOIN

Navigator: SCHEMAS

Filter objects

- bankdb
- bookflow_db
- bookflow_week2_db
- college_yt
- practice
- sql_cdg
- sys

WEEK1* ddl WEEK2 SQL File 5* SQL File 6* SQL File 7* SQL File 8*

47 LEFT JOIN class_info

48 ON class.id = class_info.id

49 WHERE class_info.id IS NULL;

50 SELECT *

51 FROM class

52 RIGHT OUTER JOIN class_info

53 ON class.id = class_info.id;

Result Grid

	id	name	id	address
▶	1	Sai Akhil	1	HYDERABAD
	2	Adhvith	2	DELHI
	NULL	NULL	3	CHENNAI
	NULL	NULL	7	BENGALURU
	NULL	NULL	8	PUNE

Result 7 ×

Output

Action Output

#	Time	Action	Message
✓ 11	20:49:50	SELECT * FROM class NATURAL JOIN cl...	2 row(s) returned
✓ 12	20:50:21	INSERT INTO class_info VALUES (7, 'BEN...	2 row(s) affected Records: 2 Duplicates: 0 ...
✓ 13	20:50:25	INSERT INTO class VALUES (5, 'Nikhil')	1 row(s) affected
✓ 14	20:52:04	SELECT * FROM class LEFT OUTER JOIN...	4 row(s) returned
✓ 15	20:52:51	SELECT * FROM class LEFT JOIN class_in...	2 row(s) returned
✓ 16	20:53:12	SELECT * FROM class RIGHT OUTER JOI...	5 row(s) returned

Administration Schemas

Information

No object selected

Chart Info Session

12.UNION

SCHEMAS

Filter objects

bankdb

bookflow_db

bookflow_week2_db

college_yt

practice

sql_clg

sys

Administration

SCHEMAS

Information

No object selected

Object Info

Session

Limit to 1000 rows

43

LEFT OUTER JOIN class_info

44

ON class.id = class_info.id;

45

• SELECT *

46

FROM class

47

LEFT JOIN class_info

48

ON class.id = class_info.id

49

WHERE class_info.id IS NULL;

50

• SELECT *

51

FROM class

52

RIGHT OUTER JOIN class_info

53

• ON class.id = class_info.id;CREATE TABLE first_table (

54

id INT,

55

name VARCHAR(30)

56

);

57

58

• CREATE TABLE second_table (

59

id INT,

60

name VARCHAR(30)

61

);

Output

Action Output

#	Time	Action	Message
✓ 13	20:50:25	INSERT INTO class VALUES (5, 'Nikhil')	1 row(s) affected
✓ 14	20:52:04	SELECT * FROM class LEFT OUTER JOIN...	4 row(s) returned
✓ 15	20:52:51	SELECT * FROM class LEFT JOIN class_in...	2 row(s) returned
✓ 16	20:53:12	SELECT * FROM class RIGHT OUTER JOI...	5 row(s) returned
✓ 17	20:53:34	CREATE TABLE second_table (id INT, ...	0 row(s) affected
✓ 18	20:53:39	CREATE TABLE first_table (id INT, n...	0 row(s) affected

Navigator

SCHEMAS

Filter objects

- bankdb
- bookflow_db
- bookflow_week2_db
- college_yt
- practice
- sql_clg
- sys

Administration Schemas

Information

No object selected

Object Info Session

SQL File 5 SQL File 6 SQL File 7

Limit to 1000 rows

```
50 • SELECT *
51 FROM class
52 RIGHT OUTER JOIN class_info
53 ON class.id = class_info.id; CREATE TABLE first_table (
54     id INT,
55     name VARCHAR(30)
56 );
57
58 • CREATE TABLE second_table (
59     id INT,
60     name VARCHAR(30)
61 );
62 • INSERT INTO first_table VALUES
63 (1, 'Akhil Karuturi'),
64 (2, 'Rohit Sharma');
65
66 • INSERT INTO second_table VALUES
67 (2, 'Sai Akhil'),
68 (3, 'MS Dhoni');
```

Output

Action Output

#	Time	Action	Message
✓ 15	20:52:51	SELECT * FROM class LEFT JOIN class_in...	2 row(s) returned
✓ 16	20:53:12	SELECT * FROM class RIGHT OUTER JOI...	5 row(s) returned
✓ 17	20:53:34	CREATE TABLE second_table (id INT, ...	0 row(s) affected
✓ 18	20:53:39	CREATE TABLE first_table (id INT, n...	0 row(s) affected
✓ 19	20:54:39	INSERT INTO first_table VALUES (1, 'Akhil...	2 row(s) affected Records: 2 Dupl
✓ 20	20:55:02	INSERT INTO second_table VALUES (2, '...	2 row(s) affected Records: 2 Dupl

Navigator: WEEK1* ddl WEEK2 SQL File 5* SQL File 6* SQL File 7*

SCHEMAS

Filter objects

- bankdb
- bookflow_db
- bookflow_week2_db
- college_yt
- practice
- sql_clg
- sys

```

66 • INSERT INTO second_table VALUES
67     (2, 'Sai Akhil'),
68     (3, 'MS Dhoni');
69
70 • SELECT * FROM first_table
71 UNION
72 SELECT * FROM second_table;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	id	name
▶	1	Akhil Karuturi
	2	Rohit Sharma
	2	Sai Akhil
	3	MS Dhoni

Administration Schemas

Information

No object selected

Result 8 x

Output

Action Output

#	Time	Action	Message
✓ 16	20:53:12	SELECT * FROM class RIGHT OUTER JOI...	5 row(s) returned
✓ 17	20:53:34	CREATE TABLE second_table (id INT, ...	0 row(s) affected
✓ 18	20:53:39	CREATE TABLE first_table (id INT, n...	0 row(s) affected
✓ 19	20:54:39	INSERT INTO first_table VALUES (1, 'Akhil...	2 row(s) affected Records: 2 Dup
✓ 20	20:55:02	INSERT INTO second_table VALUES (2, '...	2 row(s) affected Records: 2 Dup
✓ 21	20:55:17	SELECT * FROM first_table UNION SELEC...	4 row(s) returned

13.UNION ALL

File Edit View Query Database Server Tools Scripting Help

WEEK1* ddl WEEK2 SQL File 5* SQL File 6* SQL F

Limit to 1000 rows

SCHEMAS

Filter objects

- bankdb
- bookflow_db
- bookflow_week2_db
- college_yt
- practice
- sql_clg
- sys

```

70 • SELECT * FROM first_table
71 UNION
72 SELECT * FROM second_table;
73
74 • SELECT * FROM first_table
75 UNION ALL
76 SELECT * FROM second_table;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Cor

	id	name
▶	1	Akhil Karuturi
	2	Rohit Sharma
	2	Sai Akhil
	3	MS Dhoni

Administration Schemas

Information

No object selected

Result 9 ×

Output

Action Output

#	Time	Action	Message
✓ 17	20:53:34	CREATE TABLE second_table (id INT, ...	0 row(s) affected
✓ 18	20:53:39	CREATE TABLE first_table (id INT, n...	0 row(s) affected
✓ 19	20:54:39	INSERT INTO first_table VALUES (1, 'Akhil...	2 row(s) affected Re
✓ 20	20:55:02	INSERT INTO second_table VALUES (2, '...	2 row(s) affected Re
✓ 21	20:55:17	SELECT * FROM first_table UNION SELEC...	4 row(s) returned
✓ 22	20:55:46	SELECT * FROM first_table UNION ALL S...	4 row(s) returned

14.MINUS

Navigator: WEEK1* ddl WEEK2 SQL File 5* SQL File 6* SQL File 7* SQL File 8*

SCHEMAS

Filter objects

- bankdb
- bookflow_db
- bookflow_week2_db
- college_yt
- practice
- sql_clg
- sys

Administration Schemas

Information

No object selected

Object Info Session

Result Grid

id	name
1	Akhil Karuturi

Result 12

Output

Action Output

#	Time	Action	Message	D
20	20:55:02	INSERT INTO second_table VALUES (2, '...	2 row(s) affected Records: 2 Duplicates: 0 ...	0.0
21	20:55:17	SELECT * FROM first_table UNION SELEC...	4 row(s) returned	0.0
22	20:55:46	SELECT * FROM first_table UNION ALL S...	4 row(s) returned	0.0
23	20:56:10	SELECT * FROM first_table INTERSECT S...	0 row(s) returned	0.0
24	20:56:58	SELECT name FROM first_table INTERSE...	0 row(s) returned	0.0
25	20:57:38	SELECT first_table.* FROM first_table LEF...	1 row(s) returned	0.0

15.FULL OUTER JOIN

Navigator

SCHEMAS

Filter objects

bankdb

bookflow_db

bookflow_week2_db

college_yt

practice

sql_clg

sys

WEEK1*

ddl

WEEK2

SQL File 5*

SQL File 6*

SQL File 7*

SQL File 8* x

Limit to 1000 rows

94

UNION

95

96

SELECT *

97

FROM class

98

RIGHT JOIN class_info

99

ON class.id = class_info.id

100

WHERE class.id IS NULL;

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	id	name	id	address
▶	4	Kiran	NULL	NULL
	5	Nikhil	NULL	NULL
	NULL	NULL	3	CHENNAI
	NULL	NULL	7	BENGALURU
	NULL	NULL	8	PUNE

Administration

Schemas

Information

No object selected

Object Info

Session

Result 13 x

Output

Action Output

#	Time	Action	Message	Durat
✓ 21	20:55:17	SELECT * FROM first_table UNION SELEC...	4 row(s) returned	0.000
✓ 22	20:55:46	SELECT * FROM first_table UNION ALL S...	4 row(s) returned	0.000
✓ 23	20:56:10	SELECT * FROM first_table INTERSECT S...	0 row(s) returned	0.000
✓ 24	20:56:58	SELECT name FROM first_table INTERSE...	0 row(s) returned	0.000
✓ 25	20:57:38	SELECT first_table.* FROM first_table LEF...	1 row(s) returned	0.000
✓ 26	20:58:26	SELECT * FROM class LEFT JOIN class_in...	5 row(s) returned	0.000